

The empirical recurrence for OEIS A254919, recovered from its tail

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Abstract

OEIS A254919 records “Empirical recurrence of order 62” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 1646$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 17$, so the recurrence is proved for every $n \geq 79$. Its last coefficient is $c_{62} = -9895604649984 \neq 0$, so no further tail satisfies anything shorter; any order-62 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A254919 is “Number of $(n+2) \times (5+2)$ 0..1 arrays with every 3×3 subblock sum of the two maximums of the central row and central column plus the two minimums of the diagonal and antidiagonal nondecreasing horizontally and vertically.”. It has offset 1 and begins

294960, 3788492, 34618330, 125133209, 549282376, 2236028616, 9583362322, 44653950736, ...

The entry states:

Empirical recurrence of order 62 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 15:03:14, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 1646$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1646$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 62: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 17$ is the first whose minimal order is exactly the stated 62.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 3292$ exact terms from $n = 17$ on, it returns order 62 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{62} c_i a(n-i),$$

c_1	10	c_{17}	547261626	c_{33}	−23854021051440	c_{49}	−7820539854848
c_2	−6	c_{18}	−633755638	c_{34}	4298060573920	c_{50}	−547860340801536
c_3	−181	c_{19}	−2455006369	c_{35}	38925028508928	c_{51}	62524506505216
c_4	265	c_{20}	7146505469	c_{36}	105395959075072	c_{52}	558677215936512
c_5	860	c_{21}	12997747740	c_{37}	−49229564337152	c_{53}	−128402778488832
c_6	4064	c_{22}	−25048786680	c_{38}	−232680018124800	c_{54}	−136660222214144
c_7	−28137	c_{23}	−147317376749	c_{39}	−1415477920768	c_{55}	79601850122240
c_8	28616	c_{24}	108686549554	c_{40}	197556764039168	c_{56}	−157338073825280
c_9	98404	c_{25}	651235942360	c_{41}	67520728584192	c_{57}	−11001558728704
c_{10}	−227268	c_{26}	262436744816	c_{42}	166861731225600	c_{58}	203697413947392
c_{11}	−903682	c_{27}	−2134865589352	c_{43}	−97700174331904	c_{59}	−68444598829056
c_{12}	882306	c_{28}	−3198753876584	c_{44}	−602939188903936	c_{60}	−75866302316544
c_{13}	17863628	c_{29}	4808712923116	c_{45}	92888396464128	c_{61}	54425825574912
c_{14}	−29689088	c_{30}	8693986529416	c_{46}	543308269223936	c_{62}	−9895604649984
c_{15}	−55430170	c_{31}	2931581004720	c_{47}	−14138020724736		
c_{16}	−2054041	c_{32}	−19167105578368	c_{48}	−3236346462208		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 79$, and it is the recurrence OEIS A254919 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 17$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{62} - c_1 x^{62-1} - \dots - c_{62}$. Its constant term is $-c_{62} = 9895604649984$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the

components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 62 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 62, so $\deg q = 62$, and it is monic. It holds from some index t on. From $\max(t, 17)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 62, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 62, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -9895604649984 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-62 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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